

```
%oldchk=/groups/rmaier/Ravindu/Corrections/PbRhaC14C14C8T1/PbRhaC14C14T1.chk
%nprocshared=64
%mem=800GB
%chk=PbRhaC14C14T2.chk
# opt=(loose,restart) freq apfd/gen scrf=(solvent=water) guess=read pseudo=read
scf=xqc
```

PbRhaC14C14T1

0 1

Pb	1.08987700	3.94389200	-0.69394700
O	-1.52357500	1.15536800	-0.44743200
O	0.77300700	1.72512700	-1.56457400
O	-1.60991100	3.63422000	-1.75624400
C	-5.07272700	2.90374200	-0.60450100
O	-5.00301500	1.54742000	-1.07548200
C	-3.84891900	0.88755900	-0.59099900
C	-2.60483300	1.41737100	-1.31144500
C	-2.82413800	2.90811100	-1.63916900
C	-3.67899800	3.55115800	-0.56154300
O	0.94354700	3.20509500	-3.19118800
C	0.77774800	2.02965300	-2.80717100
C	0.52828600	0.96949200	-3.85305600
C	0.58032100	-0.49598300	-3.45302900
O	-0.50273000	-0.87527800	-2.56157700
C	1.82269300	-0.95971500	-2.71921300
C	3.15921300	-0.54241100	-3.32130900
C	4.26623600	-1.44111100	-2.77665600
C	5.68427300	-0.92384500	-2.97092300
C	6.71850400	-1.98564300	-2.61509100
C	8.14557500	-1.47242900	-2.49705100
C	-1.70607700	-1.11493200	-3.08379400
O	-2.02190300	-0.84145000	-4.22214700
C	-2.60650900	-1.80794200	-2.09673500
C	-3.99004100	-1.16356700	-1.89293200
O	-4.01587400	-0.47920200	-0.63723200
C	-5.79595700	2.99481700	0.72539100
O	-3.83704300	4.93964600	-0.77254900
C	-5.06820500	-2.23047700	-1.82184000
C	-6.46421000	-1.69114300	-1.56131200
C	-7.50103800	-2.80320700	-1.49407500
C	-8.90332100	-2.30633400	-1.17399500
C	-9.93343800	-3.42664000	-1.13363200
C	-11.32405100	-2.95982600	-0.72689100
C	-12.35092800	-4.08366600	-0.70030000
O	-0.56844200	3.06504300	1.04169100
O	2.47624300	2.62465400	0.71252900
O	-0.03055500	5.44772700	1.95112100
C	1.76044500	3.27753800	4.37836800
O	0.79533900	2.22056200	4.29781800

C	0.40161500	1.94567400	2.96650500
C	-0.49995700	3.07075200	2.45651500
C	0.04528200	4.43781500	2.94320500
C	1.51561500	4.34914400	3.31298100
O	3.49604100	3.29363200	-1.12462800
C	3.51122400	2.69746800	-0.01970500
C	4.77484600	2.04563800	0.48769800
C	4.64314800	0.53642500	0.66164000
O	3.71386500	0.27954900	1.74305200
C	5.98356200	-0.09599500	0.98529600
C	5.94752800	-1.61140700	1.10583700
C	7.34332700	-2.18998300	1.29603700
C	7.39700000	-3.70781000	1.20901300
C	8.79734400	-4.25809900	1.44569200
C	8.91928600	-5.74858900	1.16273600
C	2.69009600	-0.54536100	1.50894600
O	2.52630200	-1.16070200	0.47969600
C	1.73704100	-0.62602300	2.66669000
C	0.28423400	-0.31831500	2.21998400
O	-0.29667100	0.74460600	2.97803600
C	3.17708200	2.73735100	4.30567400
O	1.99229400	5.57588900	3.83912600
C	-0.62867400	-1.52044200	2.37858100
C	-2.09221600	-1.22682300	2.07794300
C	-2.97448800	-2.47023700	2.11515200
C	-4.45808000	-2.12884000	2.18330600
C	-5.37691800	-3.34192100	2.14208300
C	-6.84604200	-2.96910600	2.29225700
C	-7.78952900	-4.16371400	2.27953700
C	9.14032300	-2.59218900	-2.21782800
C	10.31542600	-6.29722900	1.42368200
H	0.28933500	-0.01698900	1.16627100
H	4.22610500	0.08849000	-0.24206700
H	-4.21719100	-0.46597500	-2.70650500
H	0.44625800	-1.07276800	-4.37332400
H	-5.63674400	3.43606100	-1.37562500
H	-3.72267000	1.10076000	0.47852600
H	-2.46025900	0.89294000	-2.25811100
H	-3.37272300	2.97007800	-2.58442700
H	-3.20856400	3.38624500	0.41810300
H	-0.44763100	1.18830800	-4.29818300
H	1.25475600	1.13384300	-4.65363900
H	1.77175200	-0.60683900	-1.68912900
H	1.77436500	-2.05374400	-2.67286200
H	3.14036700	-0.60510800	-4.41660600
H	3.36906300	0.50611100	-3.07636600
H	4.09087300	-1.60752900	-1.70688900
H	4.17333000	-2.42997400	-3.24477300
H	5.83103500	-0.59720600	-4.00851200
H	5.83568000	-0.03057200	-2.35037400

H	6.43787200	-2.45723100	-1.66538200
H	6.67892300	-2.78788900	-3.36371400
H	8.43277800	-0.94437800	-3.41577800
H	8.19701800	-0.72564000	-1.69342800
H	-2.73354700	-2.81888400	-2.49852200
H	-2.11253900	-1.90050200	-1.12987400
H	-5.84194500	4.03730200	1.05045500
H	-5.27702000	2.42291700	1.49952100
H	-6.81492600	2.61310500	0.63712200
H	-2.95555600	5.29156700	-0.94970100
H	-5.05190100	-2.79024300	-2.76425200
H	-4.79163600	-2.94147400	-1.03277200
H	-6.46340800	-1.12589600	-0.62419000
H	-6.73682300	-0.97699800	-2.34895800
H	-7.51666200	-3.34889100	-2.44695500
H	-7.19608300	-3.53509100	-0.73489200
H	-8.89129500	-1.78391300	-0.20820700
H	-9.20637800	-1.55607300	-1.91630700
H	-9.98388100	-3.90880300	-2.11893500
H	-9.59429600	-4.20436300	-0.43773200
H	-11.27537000	-2.49125000	0.26498500
H	-11.66130800	-2.17215700	-1.41387400
H	-11.99175500	-4.88825800	-0.04601100
H	-12.43581900	-4.52493600	-1.70247800
H	1.58067800	3.74481900	5.35109300
H	1.27831400	1.88040100	2.31469300
H	-1.50592100	2.91693900	2.86267600
H	-0.51648900	4.74214600	3.83409700
H	2.08119600	4.09524400	2.40841100
H	5.58417300	2.24428300	-0.21570500
H	5.03738300	2.49777000	1.44938400
H	6.67604700	0.18559800	0.18414000
H	6.37754700	0.35120500	1.90624900
H	5.30337100	-1.90942200	1.94089500
H	5.49119200	-2.03553400	0.20458800
H	8.01359400	-1.77008000	0.53482400
H	7.74677100	-1.86050500	2.26240000
H	6.69923200	-4.14974300	1.93206300
H	7.04298700	-4.02388100	0.21850400
H	9.51109900	-3.71194900	0.81524000
H	9.09880200	-4.05335600	2.48133800
H	8.19073600	-6.29885700	1.77298700
H	8.64287200	-5.94036100	0.11720200
H	1.79240300	-1.64716600	3.05475500
H	2.04249100	0.04286600	3.47072800
H	3.89183300	3.56055500	4.38567100
H	3.34579900	2.22870800	3.35394100
H	3.36559100	2.03458800	5.12018900
H	1.80544300	6.25586400	3.18165300
H	-0.25056400	-2.30613200	1.71341600

H	-0.52873400	-1.90538800	3.40195900
H	-2.46237800	-0.50231900	2.81093600
H	-2.17895700	-0.74172200	1.10233900
H	-2.77646100	-3.09150900	1.23124200
H	-2.70666300	-3.08846000	2.98329600
H	-4.64815500	-1.56785600	3.10875700
H	-4.70874000	-1.45562200	1.35792800
H	-5.23061800	-3.88414900	1.19809500
H	-5.09770900	-4.04529500	2.93848200
H	-6.98066600	-2.41534600	3.23125200
H	-7.12797000	-2.27283300	1.49202700
H	-7.48512200	-4.87783000	3.05663700
H	-7.69293300	-4.70004100	1.32591900
H	9.18107600	-3.26429000	-3.08527600
H	8.77161200	-3.20517000	-1.38548100
H	10.58800800	-6.11828100	2.47242900
H	11.04539300	-5.73660700	0.82439000
H	-1.04865100	2.27695300	0.68740000
H	-0.95194000	5.58442300	1.70660300
H	-1.34646900	3.67191300	-2.68214100
H	-0.68082400	1.19499200	-0.95627700
C	-13.72709300	-3.63215300	-0.22854000
H	-14.08588200	-2.82415500	-0.87858300
H	-13.63727400	-3.19547700	0.77419600
C	-14.74439600	-4.76601300	-0.20611900
H	-14.87328500	-5.19847700	-1.20345400
H	-14.41978600	-5.57125000	0.46067400
C	-9.24663500	-3.77522100	2.49523600
H	-9.34035600	-3.24926100	3.45369500
H	-9.54291100	-3.05224400	1.72726200
C	-10.19105800	-4.96957900	2.47137700
H	-10.11289800	-5.51549500	1.52619500
H	-9.95422300	-5.67405000	3.27512400
C	10.54365000	-2.10003900	-1.89133100
H	10.89515100	-1.44237400	-2.69629800
H	10.50434100	-1.47756400	-0.98829200
C	11.53357000	-3.24052600	-1.68961600
H	11.62201400	-3.84819900	-2.59572300
H	11.21027300	-3.90485600	-0.88231700
C	10.44869900	-7.78288100	1.11428000
H	9.71989800	-8.34308900	1.71348600
H	10.17600700	-7.95822400	0.06597700
C	11.85109000	-8.31696700	1.37740200
H	12.59238000	-7.79152300	0.76688400
H	12.13367200	-8.17980000	2.42604200
C	11.96144426	-9.81909278	1.05640344
H	11.72233874	-9.98218720	0.02629240
H	11.27861309	-10.36647687	1.67204317
C	13.39869794	-10.29866011	1.33193388
H	13.63848901	-10.13376473	2.36159876

H	13.47487096	-11.34277802	1.11076552
H	14.08134478	-9.75264326	0.71487724
C	12.93974483	-2.72360517	-1.33315394
H	13.29683054	-2.09077793	-2.11859615
H	12.89453338	-2.16775568	-0.41997966
C	13.89705329	-3.91749030	-1.16053065
H	13.93893657	-4.47581278	-2.07235399
H	14.87506780	-3.55814486	-0.91710003
H	13.54229809	-4.54803053	-0.37219838
C	-11.66055021	-4.54098262	2.64018983
H	-11.77847724	-4.03142781	3.57364959
H	-11.93548025	-3.88667177	1.83944594
C	-12.56462137	-5.78748274	2.61799093
H	-12.28984857	-6.44168180	3.41888013
H	-13.58565390	-5.48967174	2.73504301
H	-12.44651471	-6.29716910	1.68462571
C	-16.12583439	-4.28717748	0.27755912
H	-16.48582026	-3.51944268	-0.37504601
H	-16.04120384	-3.89979925	1.27137790
C	-17.11096639	-5.47084127	0.27014453
H	-18.07086497	-5.13808568	0.60595466
H	-17.19539462	-5.85838784	-0.72362583
H	-16.75111410	-6.23846572	0.92295311

C H O 0
6-311+g(2d,p)

Pb	0	
S	4 1.00	
	591.61124370	0.22126521076D-03
	46.757232559	0.56961959130D-02
	20.746462696	-0.21374063831
	14.610796419	0.40502620616
S	2 1.00	
	20.222637612	-0.83541883299D-01
	6.4767324865	0.97910892388
S	1 1.00	
	1.6600600927	1.0000000
S	1 1.00	
	0.80431655001	1.0000000
S	1 1.00	
	0.22627039020	1.0000000
S	1 1.00	
	0.84014530665D-01	1.0000000
P	3 1.00	
	15.189102118	0.61952303583
	14.693144415	-0.72498497086
	6.8705890048	0.37680007984
P	3 1.00	
	2.2021426123	0.40196284806

		1.2209125119	0.46058131862
		0.63367559815	0.19367655397
P	1	1.00	
		0.28202837058	1.0000000
P	1	1.00	
		0.11333375666	1.0000000
P	1	1.00	
		0.43948707430D-01	1.0000000
D	6	1.00	
		61.315369628	0.33870800787D-03
		12.372195840	0.13788683942D-01
		6.9254944983	-0.75979608103D-01
		2.3319539939	0.28113784298
		1.2108730003	0.44474512269
		0.60090478506	0.35326874351
D	1	1.00	
		0.28135869813	1.0000000
D	1	1.00	
		0.11500000000	1.0000000
F	1	1.00	
		1.0000000	1.0000000
F	1	1.00	
		0.2896200	1.0000000

PB	0		
PB-ECP	3	60	
f potential			
	2		
2		3.88751200	12.20989200
2		3.81196300	16.19029100
s-f potential			
	4		
2		12.29630300	281.28549900
2		8.63263400	62.52021700
2		3.88751200	-12.20989200
2		3.81196300	-16.19029100
p-f potential			
	6		
2		10.24179000	72.27689700
2		8.92417600	144.59108300
2		6.58134200	4.75869300
2		6.25540300	9.94062100
2		3.88751200	-12.20989200
2		3.81196300	-16.19029100
d-f potential			
	6		
2		7.75433600	35.84850700
2		7.72028100	53.72434200
2		4.97026400	10.11525600

2	4.56378900	14.83373100
2	3.88751200	-12.20989200
2	3.81196300	-16.19029100